Sample Weights

Overlapping outcomes

In financial ML we assign a label y_i from features X_i, where y_i was a function of price bars occuring between $[t_{i,0}; t_{i,1}]$.
(trade entry; trade stop)

⚠ Problem: When $t_{j,0} < t_{i,1}$, y_j and y_i depend on a common return $r_{t_{j,0};\min\{t_{j,1}, t_{i,1}\}}$. Therefore series of labels is NOT IID.

Label uniqueness

We are determining how unique is a sample to derive a sample weight **for training phase**.
(how much is this unique information equivalent to how much model should learn from it)

1. Instantaneous uniqueness $(u_{t,i})$
   - Find concurrency : $c_t = \sum_{i=1}^{I} 1_{t,i}$ (sum active labels at time t)
   - Uniqueness at time t : $u_{t,i} = \dfrac{1_{t,i}}{c_t}$
2. Average uniqueness $\bar{u}_i$: $\dfrac{\sum_{t_{i,0}}^{t_{i,1}} u_{t,i}}{\sum_{t_{i,0}}^{t_{i,1}} 1_{t,i}}$

Bagging classifiers and uniqueness

In standard bootstrapping (IID case) where you sample I items with replacement:

- Probability of not selecting one item in one draw: $1 - \frac{1}{I}$
- Probability of never selecting: $\lim_{I \to \infty} (1 - \frac{1}{I})^I = e^{-1} \approx 36.8\%$

In overlapping bootstrapping, there are only K chunks of unique information where $K \ll I$.

- Probability becomes $(1 - \frac{1}{k})^I = ((1 - \frac{1}{k})^k)^{I/k} \approx e^{-I/k}$

With $k \leq I$, ratio I/k becomes larger and therefore we will end up selecting every sample.

⚠ Problem: heavy oversampling of same data, destroying performance of bagging classifiers like Random Forest.

Sequential Bootstrapping

Solution can be to bootstrap based on dynamic probabilities instead of uniform distribution

1. Initial draw: $\delta_i^{(1)} = \frac{1}{I}$, $\forall_i = 1, \dots, I$, draw becomes $\phi^{(1)} = \{i$

2. Before next draw we evaluate instantaneous uniqueness of every sample if added to timeline $\phi^{(1)}$:

$$u_{t,j}^{(2)} = \frac{1_{t,j}}{1 + \sum_{k \in \phi^{(1)}} 1_{t,k}}$$

3. Then convert to valid probability distribution:

$$\delta_j^{(2)} = \frac{\bar{u}_j^{(2)}}{\sum_{k=1}^{I} \bar{u}_k^{(2)}}$$

$$\{1_{t,i}\} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$ say $\phi^{(1)} = \{2\}$

For y_1 : $u_{1,1} = 1$, $u_{2,1} = 1$, $u_{3,1} = \frac{1}{2}$
↳ $\bar{u}_1^{(2)} = \frac{1 + 1 + 1/2}{3} = 5/6$

For y_2 : $u_{3,2} = 1/2$, $u_{4,2} = 1/2$ (repeats itself)
↳ $\bar{u}_2^{(2)} = \frac{1/2 + 1/2}{2} = 1/2$

For y_3 : $\bar{u}_3^{(2)} = 1$ => $\delta^{(2)} = \{\frac{5}{14}; \frac{3}{14}; \frac{6}{14}\}$

Therefore repetition is restricted (smaller probabity of picking same item). Proving closer to IID behavior.

Return Attribution

Overlap problem: solve overlapping labels problem with uniqueness weighting.
Return problem: trade signal with 10% move >> important than signal with 0.5% move.

1. Compute instantaneous uniqueness c_t
2. Attribute return of a bar:
   - If i has $c_t = 1$ (unique), get $R_i = 1 \times \sum_{t_{i,0}}^{t_{i,1}} r_t$
   - Otherwise $R_i = \sum_{t_{i,0}}^{t_{i,1}} r_t / c_t$
3. Determine sample weight $w_i = |R_i|$ and scale so that sum equal number of obs. I.
   (only works under binary classification)

Time Decay

Because market are adaptive systems, older data becomes less reflective of current market regimes
We can multiply return weights by a time decay factor d.

- $d = 1$, no time decay => recent data.
- $0 < d < 1$, downward slope with time.

⚠ How is time measured?
"Time" only ticks forward when unique information comes in.
Therefore if market has highly redundant information, time decay slows down.

Sklearn Class Weights

If data contains 99% normal days, 1% market crash, model get 99% free accuracy.

- Use class_weight = 'balanced'
  ↳ where $w_j = \dfrac{\text{Total samples}}{\text{number of classes} \times \text{count of class } j}$

- Final weight$_i$ = Sample Weight$_i$ × Class Weight$_i$